PUSL3190 Computing Project

Project Proposal

Pomegranate Ripeness and Disease Detection System

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Chapter 01 - Problem Statement

Fruit quality assessment is a critical process in agriculture, in developing countries like Sri Lanka determining fruit ripeness and detecting early signs of disease primarily rely on manual inspection methods, which are time consuming and prone to human error. Farmers, particularly those in rural regions with limited access to advanced technologies often experience significant losses due to improper harvesting times, poor storage decisions or undetected diseases that spread rapidly within crops.

Although recent advancements in computer vision and machine learning have demonstrated great potential in automating agricultural monitoring, most existing systems depend on cloud based processing or high-end hardware, both of which are often unaffordable and impractical for small scale farmers. Limited internet connectivity and the high computational requirements of these systems make them unsuitable for deployment of standard mobile devices commonly used in rural farming communities.

Moreover, for crops such as pomegranate which is gaining popularity in Sri Lanka through newly introduced local varieties like Lanka Red and Malee Pink there is a lack of localized datasets and automated tools capable of accurately identifying ripeness stages and detecting diseases specific to these cultivars. This absence of region-specific data further reduces the accuracy of machine learning models.

Therefore, there is an urgent need for a lightweight, real time solution that enables farmers to assess fruit quality effectively without relying on constant internet connectivity or expensive infrastructure. The proposed system aims to address these challenges by developing a mobile application integrated with optimized machine learning models. Additionally, by incorporating an augmented reality (AR) interface the system will provide interactive visual feedback on ripeness and disease status directly on the live camera feed, enabling farmers to gain deeper insights into their fields and make more informed decisions.

Chapter 02 - Project description

2.1 Background and Rationale

Agriculture plays a critical role in Sri Lanka's economy with fruit cultivation contributing substantially to both local food production and export value. Among these crops pomegranate (*Punica granatum*) has emerged as a high value fruit due to its nutritional benefits and market demand. However, determining the correct harvest time and detecting early stage diseases remain as challenges for farmers. Even though manual inspection is traditional they are inefficient and prone to error particularly under diverse environmental conditions.

2.2 Project Overview

This project aims to design and implement a real-time pomegranate fruit ripeness and disease detection system using lightweight on device ML models integrated into a Flutter based mobile application. The system is designed to be efficient, accessible and user friendly and specifically adaptable to low resource agricultural environments.

Lightweight detection architectures such as YOLOv8 and EfficientDet will be used to localize pomegranate fruits from live camera input while MobileNet or EfficientNet classifiers will assess ripeness stages (unripe, semi ripe, and ripe) and identify visible disease symptoms. These models will be optimized for TensorFlow Lite (TFLite) allowing low latency and offline inference suitable for mid range Android smartphones.

Existing studies including YOLO-Granada by Zhao et al. (2024) and YOLO-MSNet by Xu et al. (2025), have proven the capability of lightweight YOLO models for pomegranate detection and disease classification. Similarly, Al Ansari (2024) developed an AlexNet-based Android application for identifying pomegranate leaf diseases directly from mobile captured images, demonstrating the practicality of mobile based agricultural AI systems. However, these studies have primarily used datasets and environmental conditions from other regions limiting their adaptability to Sri Lankan varieties such as Lanka Red and Malee Pink.

2.3 Dataset Development

A distinguishable component of this project is the creation of a custom dataset focusing on Sri Lankan pomegranate varieties. These varieties differ from globally studied cultivars in color, texture and shape, which necessitates localized data for accurate detection. Images will be captured under various lighting, weather and background conditions to ensure model robustness and adaptability to real world field environments.

To enhance dataset diversity image augmentation techniques including rotation, scaling, brightness adjustment, flipping will be applied. Additionally, publicly available datasets from Kaggle will be used to complement the dataset. This hybrid approach helps the model generalize better across varying fruit appearances while maintaining high accuracy in Sri Lankan conditions.

2.4 Scope of the Project

Included

- On-device ML models for fruit detection, ripeness classification, and disease identification.
- Dataset development, augmentation, and preprocessing.
- Flutter mobile app with real-time camera inference.
- TensorFlow Lite optimization and performance benchmarking.
- AR-based visualization overlays on live camera feed.

Excluded

- Cloud-based inference or server side processing.
- Disease detection beyond visually observable symptoms.

2.5 Expected Outcomes

- A fully functional mobile application with offline inference capability
- Optimized ML models suitable for on device deployment
- Custom dataset representing Sri Lankan pomegranate varieties
- AR enhanced visual feedback for ripeness and disease information
- Benchmark analysis for accuracy, frame rates and latency on mid-range Android devices

2.6 Advantages and Practical Impact

Unlike conventional cloud based agricultural applications, the proposed solution performs all computations entirely on device ensuring offline operation even in areas with poor or no internet connectivity. The system's lightweight architecture and use of TensorFlow Lite optimization minimize latency while maintaining high inference accuracy.

By supporting efficient real time decision making, the proposed system contributes to reducing harvest losses, improving fruit quality assessment and promoting data driven farming practices. Ultimately, it aligns with Sri Lanka's vision of smart and sustainable agriculture.

2.7 Project Objectives

The primary objectives of this project are:

- To develop and optimize lightweight ML models for detecting pomegranate fruits, classifying ripeness, and identifying disease conditions.
- To integrate these models into a Flutter based mobile application with on device inference using TensorFlow Lite.
- To design an AR based visualization interface that overlays ripeness and disease information on the live camera feed.
- To evaluate model performance based on detection accuracy, latency and real time frame rates on mid-range Android devices.
- To deliver an affordable, offline capable mobile solution that enhances productivity, reduces waste and supports Sri Lankan farmers in low resource agricultural environments.

2.8 Project Keywords

- Mobile Machine Learning
- On-Device AI
- Precision Agriculture
- Object Detection
- Augmented Reality

Chapter 03 - Research gap

Despite rapid advancements in agricultural automation and fruit detection technologies, the domain of pomegranate ripeness and disease detection particularly in mobile based, low-resource contexts remains underexplored. While several studies have achieved high accuracy using deep learning and computer vision techniques most of these solutions are optimized for laboratory environments or rely on cloud based processing, making them impractical for small scale farmers in developing countries like Sri Lanka. Furthermore, existing systems rarely consider the variability in regional cultivars, environmental conditions, and resource limitations of mobile hardware. This chapter identifies the key research gaps in current literature and technologies, realizing the need for a localized, lightweight and offline mobile solution that integrates real time detection with user friendly visualization through augmented reality (AR).

3.1 Knowledge Gap

Although several studies have explored pomegranate detection and disease identification using computer vision and machine learning, most research focuses on global or generalized datasets. Existing models lack adaptation to regional fruit varieties such as Lanka Red and Malee Pink, which are unique to Sri Lanka. There is a lack of comprehensive studies addressing fruit ripeness and disease detection specifically for these local cultivars under Sri Lankan environmental and lighting conditions.

3.2 Technology Gap

Xu et al. (2025) introduced YOLO-MSNet (YOLOv11n), a pruned YOLO model optimized for pomegranate detection, achieving notable accuracy improvements and reduced model size for efficient deployment. Similarly, Zhao et al. (2024) developed the YOLO-Granada project, which implemented a lightweight YOLOv5/ShuffleNet model for Android to classify pomegranate growth stages and detect disease symptoms in real time. However, these systems still rely on datasets and environmental conditions from other regions, and their architectures, though efficient, have not been fine-tuned or validated for low-resource mobile environments typical of Sri Lankan farming communities. Moreover, most of these studies focus on detection accuracy rather than the user experience or integration with augmented reality (AR) interfaces that could improve usability in field applications.

3.3 Methodological Gap

Existing models primarily depend on pre-trained datasets or synthetic augmentation, which may not represent real-world variations in fruit color, texture, or disease appearance found in Sri Lanka. The absence of region-specific datasets limits the generalization ability of these models when deployed in local contexts.

3.4 Contextual Gap

No current research has developed or deployed a lightweight, on-device machine learning system tailored for Sri Lankan pomegranate varieties that can operate efficiently offline and provide interactive feedback to farmers.

To resolve these gaps this project proposes to create a localized dataset of Sri Lankan pomegranate images covering multiple ripeness stages and disease conditions to fine tune existing lightweight models such as YOLOv8 and MobileNet. The project will further integrate an augmented reality (AR) interface within a mobile application to deliver real time ripeness and disease visualization directly on the device ensuring accessibility for farmers with limited connectivity and resources.

Chapter 04 - Requirement Analysis

These requirements ensure that the proposed system remains both technically efficient and contextually relevant. By combining lightweight machine learning models with an intuitive AR based interface, the application directly addresses the identified research gaps in accessibility and localization. Its offline mobile application based design allows real time fruit detection and classification even in low resource agricultural environments supporting Sri Lankan farmers with a practical and scalable technological solution.

Here is the table regarding the Functional, non-functional and user requirements:

Type	**Description**
Functional	The system shall capture images using mobile cameras and detect pomegranate fruits in real time.
Functional	The system shall classify each detected fruit into ripeness categories and identify any diseased fruits using an augmented reality AR overlay
Functional	The system should provide more information about whether pomegranate fruit is diseased.
Non-Functional	The system should operate offline with minimal latency
Non-Functional	The application must run smoothly on mid-range Android devices.
Non-Functional	The system should have a simple, user friendly UI for users.
User	Users should be able to detect fruit into ripeness categories and identify any diseased fruits and gain insights into them.

Chapter 05 – Finance

his project does not require any external financial allocation. All development and experimentation will be conducted using my personal laptop which provides sufficient computational resources for model training and testing.

The dataset used for model fine tuning is retrieved from two sources:

- A self created dataset containing images of locally grown Sri Lankan pomegranate varieties (Lanka Red, Malee Pink, etc.).
- An existing open source dataset available on Kaggle under the “Fruits / Pomegranate” section for reference and augmentation.

Since both the hardware and datasets are available at no cost and the required software tools such as TensorFlow, PyTorch and Flutter are open source no budget or financial expenditure is required for this project.

Chapter 06 - External organizations

The project will be carried out in collaboration with Ms. L. G. I. Samanmalee, Assistant Director of Agriculture, Department of Agriculture, who will provide insights into pomegranate cultivation and supply images of pomegranates for the dataset used to train the machine learning model.

Chapter 07 - Time Frame / Timeline

The project will be carried out in alignment with the university semester schedule from November 2025 to October 2026. The Gantt chart below illustrates the planned project activities and timeline.

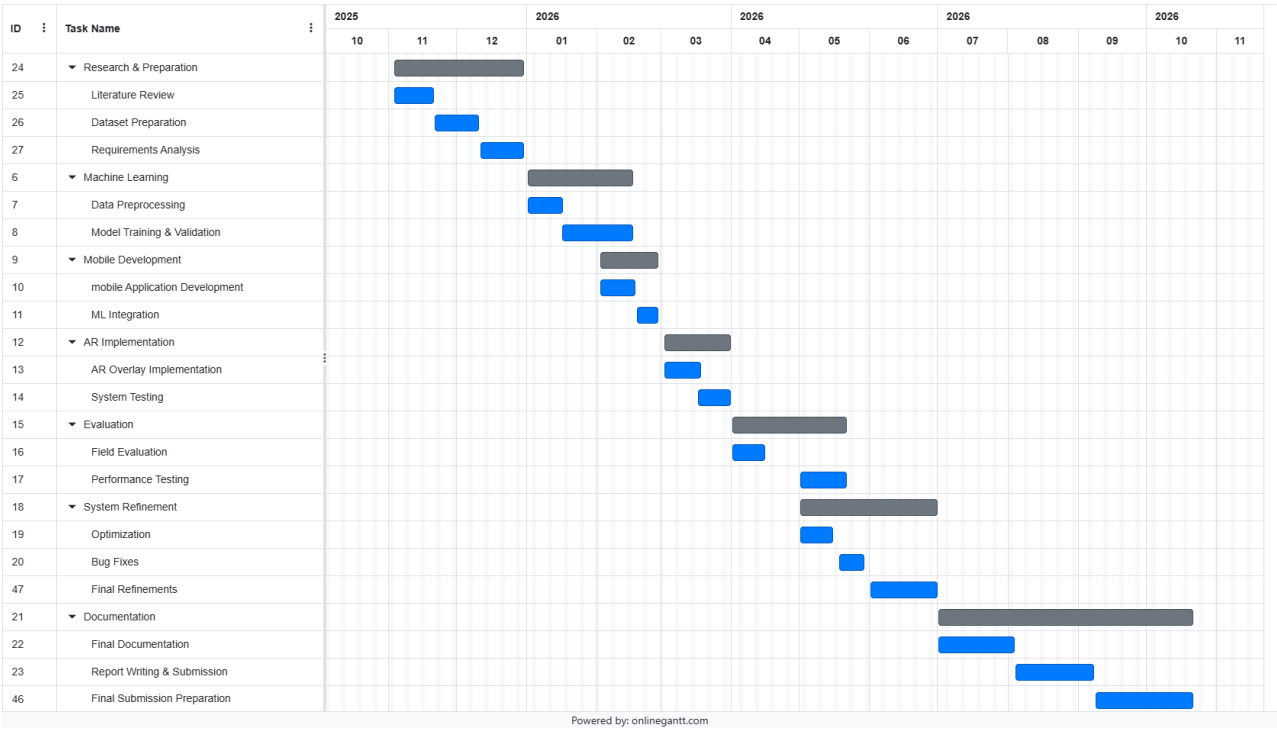

Figure: Project Gantt chart illustrating the timeline for pomegranate ripeness and disease detection system.

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